CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 22: Heights and Distances

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. The angle of elevation of the top of a pole from a point 15 m from its foot is $45°$. Find the height of the pole.

2. A ladder 8 m long reaches a point 4 m up a wall. Find the angle it makes with the ground.

3. The shadow of a vertical tower is 20 m long when the sun's altitude is $60°$. Find the height of the tower.

4. From the top of a building 30 m high, the angle of depression of a car is $30°$. Find the distance of the car from the building.

5. A man 1.6 m tall stands 20 m from a tree and sees its top at an angle of elevation of $45°$. Find the height of the tree.

6. The angle of elevation of the top of a tower is $45°$. On walking 20 m towards the tower, it becomes $60°$. Find the height of the tower.

7. Two men on opposite sides of a tower 60 m high observe its top at angles of elevation $45°$ and $60°$. Find the distance between the men.

8. From the top of a cliff 90 m high, the angles of depression of two boats on the same side, in line with the foot, are $30°$ and $60°$. Find the distance between the boats.

9. From the top of a tower 100 m high, the angles of depression of the top and bottom of a building are $30°$ and $45°$. Find the height of the building.

10. An aeroplane flying at a height of 3000 m passes vertically above another aeroplane. From a point on the ground, their angles of elevation are $60°$ and $45°$. Find the vertical distance between them.

MathSetu

Answer Key

1. 15 m
2. $\sin\theta = \frac{1}{2}$, so 30°
3. $20\sqrt{3} \approx 34.64$ m
4. $30\sqrt{3} \approx 51.96$ m
5. $20 + 1.6 = 21.6$ m
6. $h = \frac{20\sqrt{3}}{\sqrt{3}-1} = 10(3+\sqrt{3}) \approx 47.32$ m
7. $60 + 20\sqrt{3} \approx 94.64$ m
8. $90\sqrt{3} - 30\sqrt{3} = 60\sqrt{3} \approx 103.92$ m
9. $d = 100$ m; $100 - \frac{100}{\sqrt{3}} \approx 42.26$ m
10. $d = \frac{3000}{\sqrt{3}} = 1000\sqrt{3}$; lower plane at 1732 m; distance ≈ 1268 m

MathSetu